

SHIVAM OJHA

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SUMMARY

Senior Machine Learning Engineer specializing in LLM post-training, agentic AI systems, and scalable ML infrastructure. Experienced in RL-based optimization, multimodal models, and productionizing reasoning-driven AI systems.

TECHNICAL SKILLS

- Languages & Frameworks: Python, PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, LangGraph
- ML/AI Expertise: LLMs (RAG, QLoRA, fine-tuning), VLMs, Vision Models (ViT, ConvNeXt, SAM2), Multimodal AI
- MLOps & Deployment: Docker, Kubernetes, vLLM, AWS (SageMaker, Textract, Bedrock), Ray Serve
- Experimentation & Tools: SQL, Pandas, NumPy, Elasticsearch, Airflow, Matplotlib

WORK EXPERIENCE

Senior Research Engineer, LLM Core & Agentic AI | Capital One (New York, NY) Jan 2026 – Present

- Developing multi-layer performance optimizations for agentic AI systems: expert-pruning on GPT-OSS to reduce model compute, latency/throughput benchmarking (TTFT, TPOT, tokens/sec) across batch sizes to inform model-selection decisions, and system-level optimizations within vLLM.
- Built a standardized multi-agent platform enabling downstream teams to productionize agentic workflows efficiently.
- Designed and implemented a pluggable memory system for agentic workflows, drawing on recent research (FluxMem, A-MEM) to enable dynamic, self-organizing memory that adapts its structure to interaction patterns rather than relying on fixed storage and retrieval.
- Implemented modular self-reflection loops using a generator-reflector-curator design (as in ACE), where a reflector distills lessons from execution feedback and a curator consolidates them into an evolving context playbook, improving agent performance under both offline and online context adaptation.
- Conducting LLM post-training with agentic reinforcement learning to improve reasoning quality and coordination in multi-agent systems.

Senior Machine Learning Engineer, R&D | CCC Intelligent Solutions (Chicago, IL) Jun 2022 – Dec 2025

- Fine-tuned policy-constrained LLMs using RLVR with GRPO, ranking group-sampled reasoning trajectories via automated insurance policy, consistency, and grounding verifiers to improve compliance and reasoning reliability.
- Led research and development of in-house LLM finetuning and serving infrastructure, spanning dataset curation, QLoRA-based fine-tuning, evaluation harnesses, and multi-node deployments; optimized end-to-end throughput and latency using vLLM, accelerating experimentation and production readiness for large language models (LLaMA-4, Qwen).
- Developed a repair assistant agent using agentic RAG with automated retrieval evaluation, safety guardrails, and a voice interface, leveraging Whisper for speech-to-text, citation-based retrieval, and ReAct-style reasoning for grounded, auditable responses.
- Built an AI agent for real-time photo validation using VLMs with SAM2 for interactive damage segmentation, optimized multimodal inference with prefix caching and request batching, and fine-tuned Qwen 3 for NER to extract vehicle damage from voice-driven claim workflows.
- Implemented a multimodal medical coding pipeline combining a VLM for clinical note understanding and a fine-tuned LLaMA 2 model for bill validation, with rule-based and LLM guardrails to improve accuracy and review reliability.
- Designed and trained ViT-based computer vision models to estimate chrome panel damage and other part-level vehicle attributes, incorporating custom loss functions to disambiguate visually similar operations.
- Developed a self-supervised transformer, training it with a Masked Language Modeling (MLM) objective to build a deep understanding of part context, then adapting it to perform auto-regressive decoding at inference to predict latent damage with high precision.

Data Science & Cognitive Analyst | Deloitte (Bengaluru, India) Aug 2019 - Jul 2021

- Optimized RL agent with custom rewards and tuning, securing top-5 in Deloitte AWS DeepRacer Virtual League.
- Conducted rigorous EDA on medical datasets to identify key predictive features for patient outcomes, establishing performance baselines with POC models in AWS SageMaker to guide future development.

EDUCATION

Columbia University

M.S. in Electrical Engineering, Specialization in Machine Learning, CGPA: 4.09/4

New York, NY

Dec 2022

Coursework: Machine Learning, Deep Learning, Computer Vision, Algorithms, Big Data Analytics

Manipal Academy of Higher Education, Manipal Institute of Technology

B.Tech. in Electrical and Electronics Engineering, CGPA: 8.35/10

Manipal, India

May 2019

AWARDS

- Recipient of the MS EE Award of Excellence, a distinction for top achievement granted annually to less than 5% of all MS students.

PUBLICATIONS

- **Real-time Multi-Camera Analytics for Traffic Information Extraction and Visualization** – 2023 IEEE International Conference on Pervasive Computing, March 2023